文献汇报

高铭阳

2020.05.08

Real-time leak detection using infrared camera and Faster R-CNN technique

Jihao Shi[1], Yuanjiang Chang[1], Changhang Xu[1], Faisal Khan[2], Guoming Chen[1], Chuangkun Li[3]

[1] Centre for Offshore Engineering and Safety Technology, China University of Petroleum, Qingdao 266580, China

[2] Centre for Risk, Integrity, and Safety Engineering (C-RISE), Faculty of Engineering and Applied Science, Memorial University of Newfoundland, St. John's, NL A1B 3X5, Canada

[3] State Key Laboratory of Safety and Control for Chemicals, Sinopec Qingdao Research Institute of Safety Engineering, Qingdao 266580, China

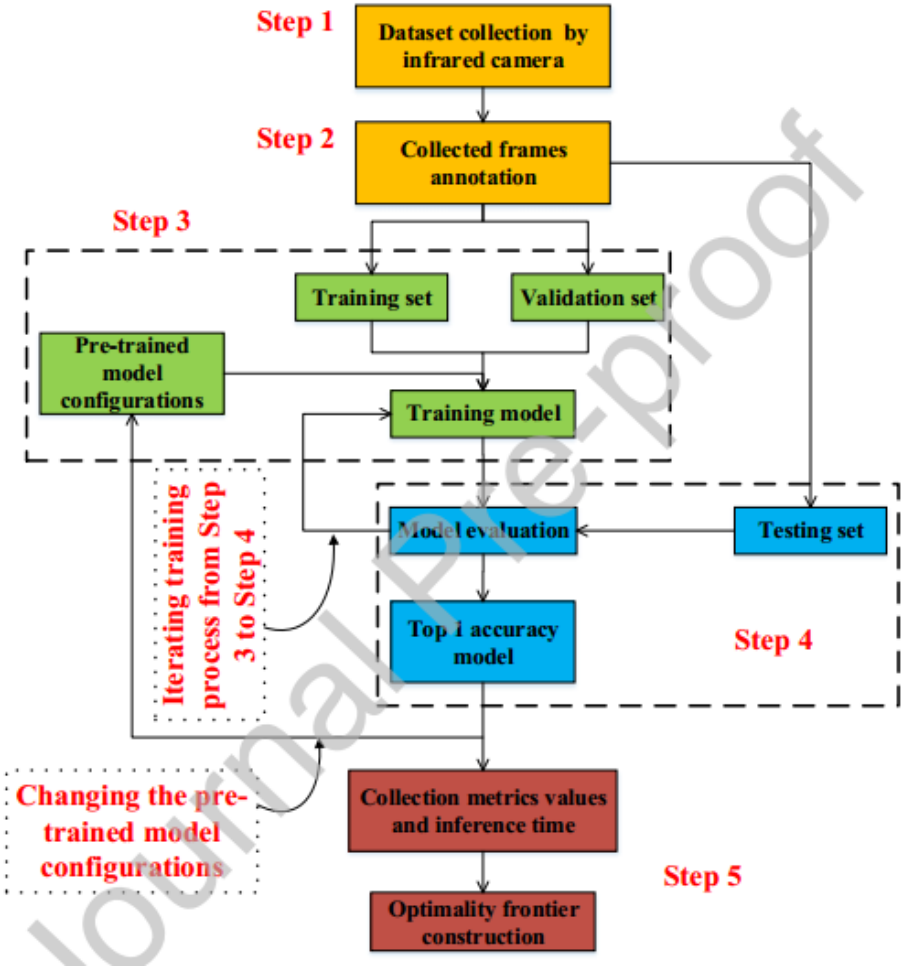
Publish：Computers and Chemical Engineering

介绍

工业火灾爆炸事故通常都是由泄漏事故所导致的。早期发现油气泄漏是预防火灾爆炸事故发生的关键。泄漏检测通常有两个任务，识别泄漏(通常称为分类任务)和定位泄漏(通常称为定位任务)。分类任务还检查烃泄漏类型，定位任务还建立烃泄漏源。这两项任务对于快速预防事故决策都是至关重要的。

检测方法

一、使用传统的传感器检测。
二、利用光学气体成像仪器，OGI的优势在于，它可以在几米之外实现实时的碳氢化合物泄漏检测。利用CNN进行识别任务，但不能定位泄漏气体。



Journal Pre-proof

Table 1 Configuration of the FLIR GF 320 infrared camera

	Hydrocarbon leaks imaging
Resolution	320×240 pixels
Focal length	38 mm
Sensor type	Stirling Microcooler
Field of View (FOV)	$14.5^{o}\times10.8^{0}$
Focus	Auto&Manual

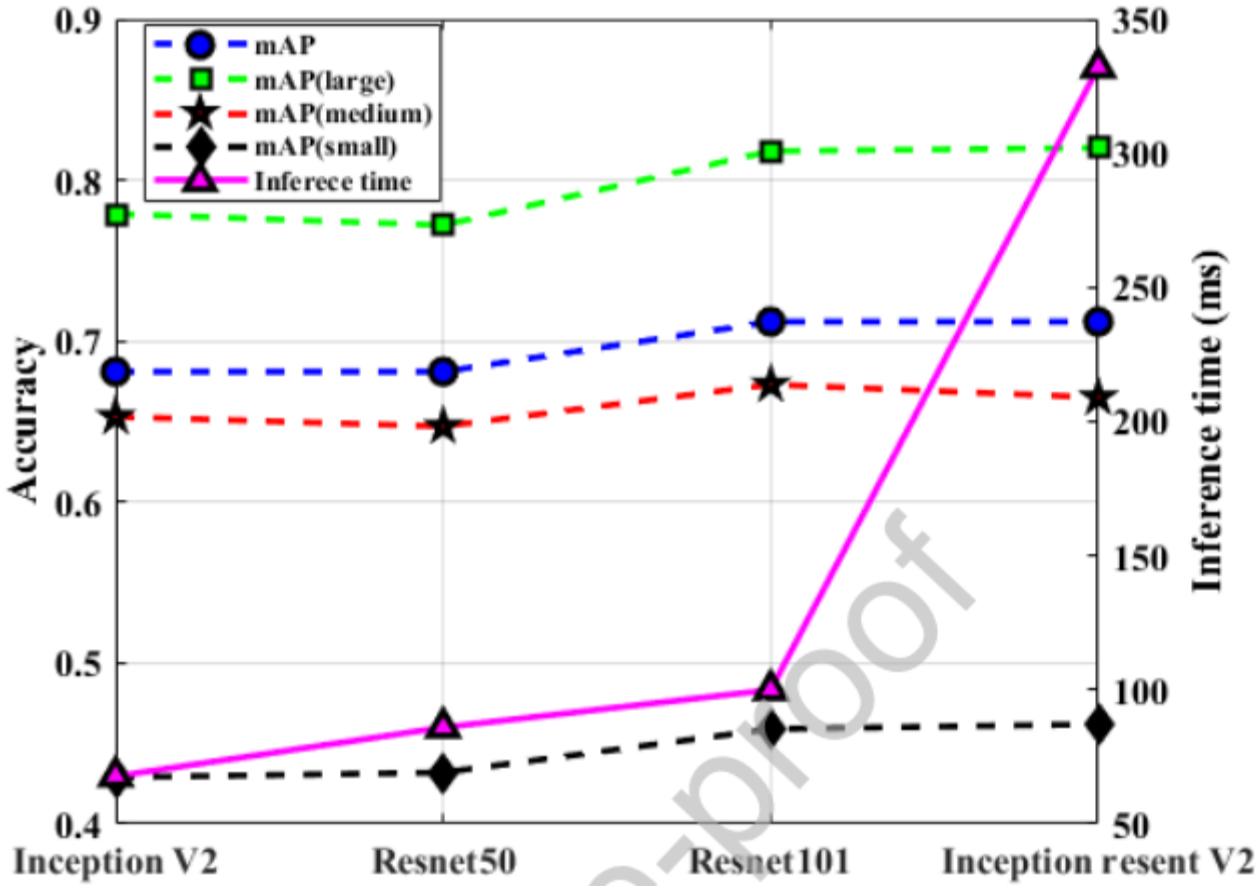


Fig. 6 Effect of feature extractor on the Faster R-CNN approach's accuracy and inference time based on a pre-trained model trained by the COCO dataset and 300 proposals. Note that the mAP@0.5 IOU values are not demonstrated to ensure the readers clearly understand the content.

Pre-proof

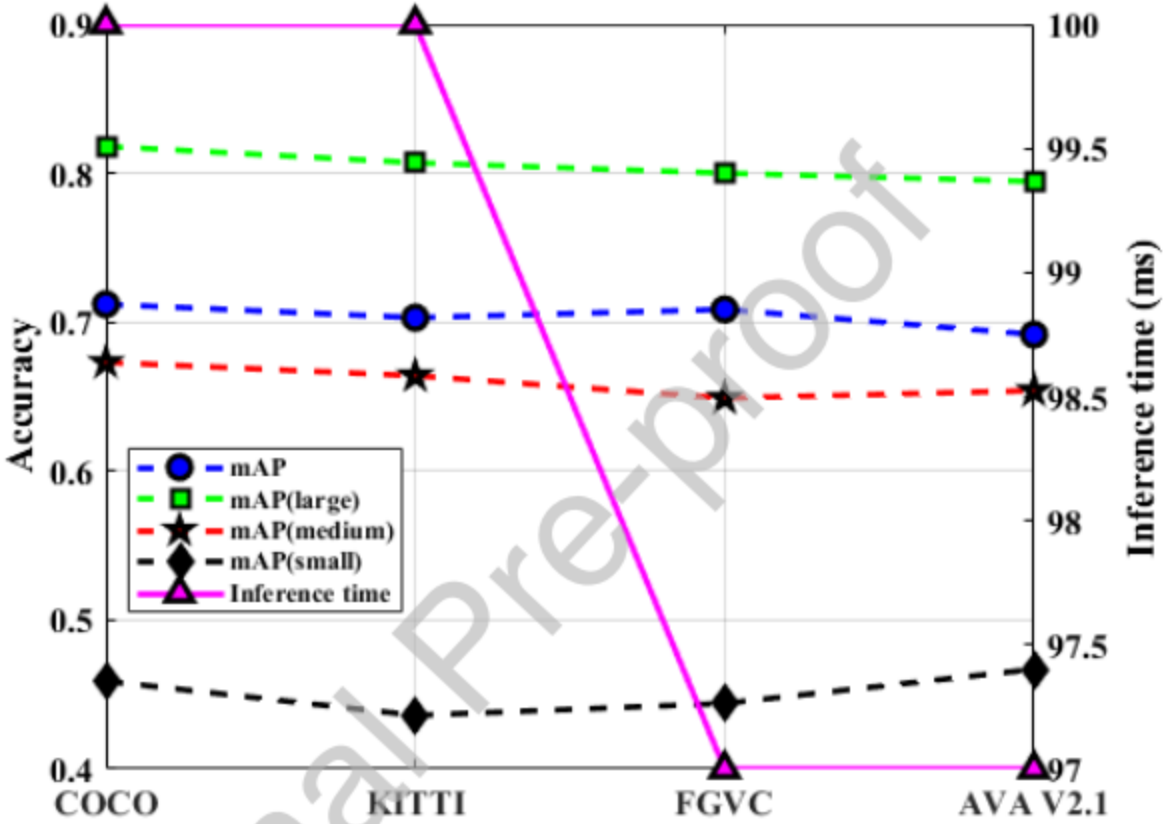


Fig. 7 Effect of the pre-trained model trained by varied datasets on the Faster R-CNN approach's accuracy and inference time under the Resnet 101 feature extractor and 300 proposals

Journal Pre-proof

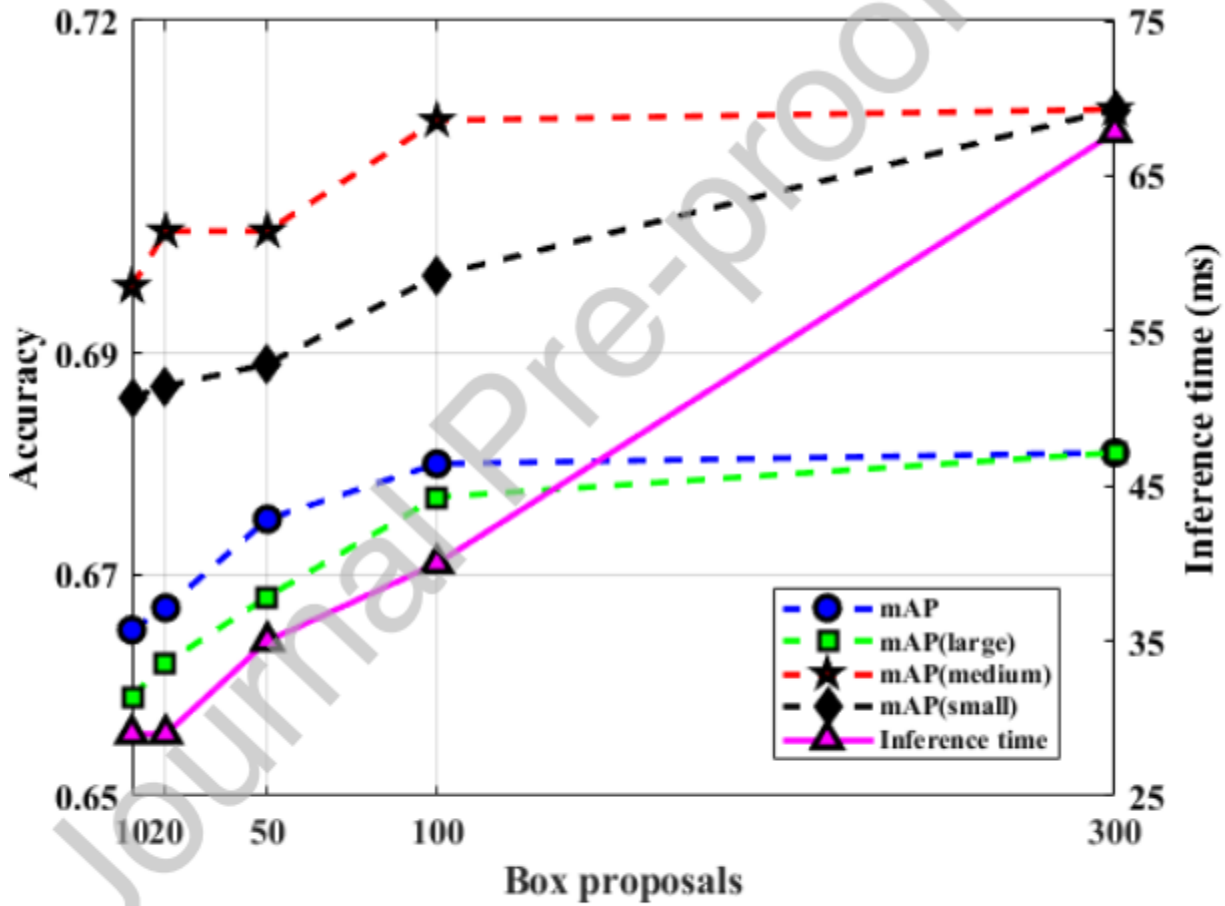
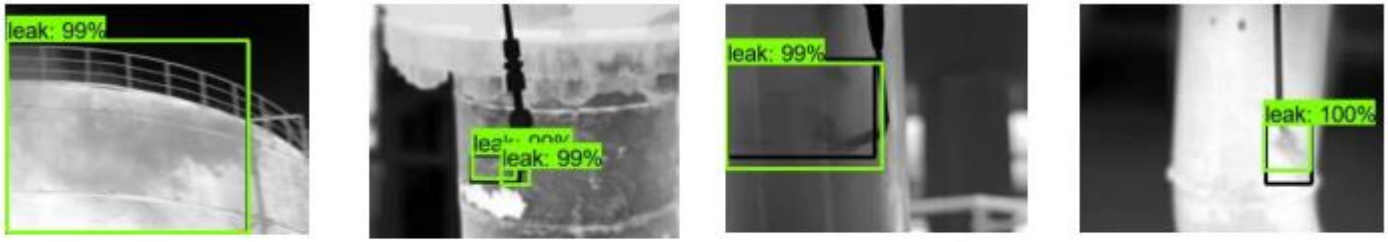


Fig. 8 Effect of the number of proposals on the Faster R-CNN approach's accuracy and inference time under the Resnet 101 feature extractor and pre-trained model by the COCO dataset

Journal Pre-proof

Table 3 Comparison between the Faster R-CNN and SSD approaches

	mAP	mAP(large)	mAP(medium)	mAP(small)	mAP@0.5 IOU	Inference time
Ours	**0.711**	**0.816**	**0.671**	**0.462**	**0.979**	**60**
SSD mobilenet v1 coco	0.637	0.756	0.589	0.320	0.942	11
SSD mobilenet v2 coco	0.675	0.799	0.612	0.351	0.967	13
SSD inception v2 coco	0.678	0.820	0.622	0.421	0.963	15



a b c d

Fig. 10 Examples visualization of the optimal Faster R-CNN model based on the Faster rcnn resnet 101 coco with 100 proposals

Table 3 Total smoke and non-smoke samples in our dataset

Video type	Number of videos	Number of clips	Number of frames
Smoke video	71	1786	107584
Simple non-smoke video	55	1304	77973
Challenge non-smoke video	31	648	38965

Table 4 Detection Rate(DR), False Alarm Rate(FAR) and Accuracy Rate(AR) of our proposed method, and comparisons with handcrafted feature-based methods and other deep learning based methods

Algorithms	DR	FAR	AR
GLCM+NN (Srisuwan et al. [24])	91.2%	10.3%	90.4%
NRLBP+SVM (Tian et al. [27])	92.1%	9.3%	91.3%
BoW+RF (Ko et al. [16])	93.9%	8.7%	92.5%
Spatial stream ConvNet (AlexNet, Tao et al. [26])	95.2%	6.2%	94.4%
Deep normalization ConvNet (Yin et al. [31])	95.9%	5.9%	95.0%
Temporal stream ConvNet	92.7%	7.9%	92.4%
Spatial-temporal stream ConvNet(Proposal 1)	95.4%	5.8%	94.8%
Enhanced spatial-temporal ConvNet(Proposal 2)	97.0%	3.5%	96.7%